# ECON 0150 | Economic Data Analysis

*Building economic models with data.*

## *Part 3.3 | Quantifying Uncertainty*

# New Jersey Minimum Wage Study

*The simplest economic model predicts employment should **fall**.*

- *NJ raised its minimum wage in 1992.*
- *Card and Krueger measured how 331 NJ restaurants' employment changed.*

- *Employment increased by **0.59 FTE workers** per store on average.*
- *This is the **sample mean** ($\bar{x}$) from $n = 331$ restaurants.*

# New Jersey Minimum Wage Study

*The simplest economic model predicts employment should **fall**.*

***Question.*** *How do we learn about the population mean ($\mu$) if all we observe is $\bar{x} = 0.59$ FTE?*

# Central Limit Theorem: Refresher

*The sample mean follows a normal distribution around the true mean ($\mu$).*

The sample mean $\bar{x}$ of (nearly) any distribution follows a normal distribution centered on the population mean $\mu$ with standard deviation $SE = \sigma/\sqrt{n}$.

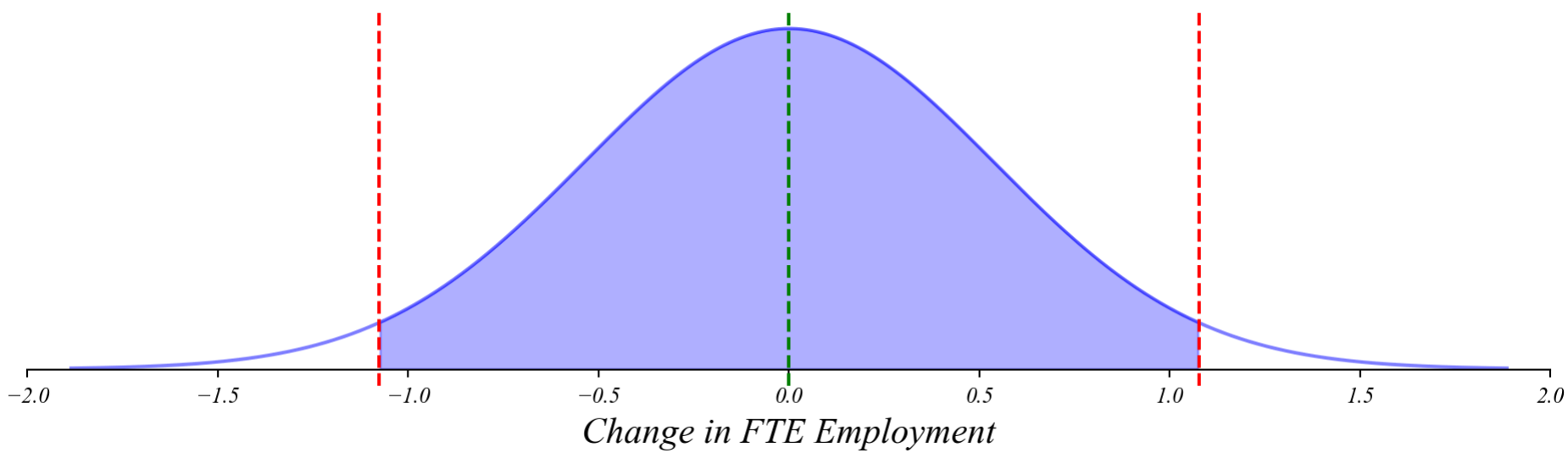
$$\bar{x} \sim N\left(\mu, \frac{\sigma}{\sqrt{n}}\right)$$

- *We don't need to know the shape of the distribution to know the shape of the sampling distribution.*
- *The sampling distribution (what we draw our sample mean from) is centered on the population mean.*
- *The variability in sample means gets smaller as the sample size grows.*

# Confidence Intervals: Known $\mu$ and $\sigma$

*We can calculate all kinds of probability if we know just $\mu = 0$ and $\sigma = 9.8$.*

**Question**. With $n = 331$, what's the probability $\bar{x}$ is within $2 \times SE$ of $\mu$?




> $P(\mu - 2 \times SE \leq \bar{x} \leq \mu + 2 \times SE) \approx 0.95$

This interval will contain 95% of our sample means.

# Exercise 3.3 | Confidence Intervals: Known $\mu$ & $\sigma$

$\sigma = 9.8$ *and* $n = 331$

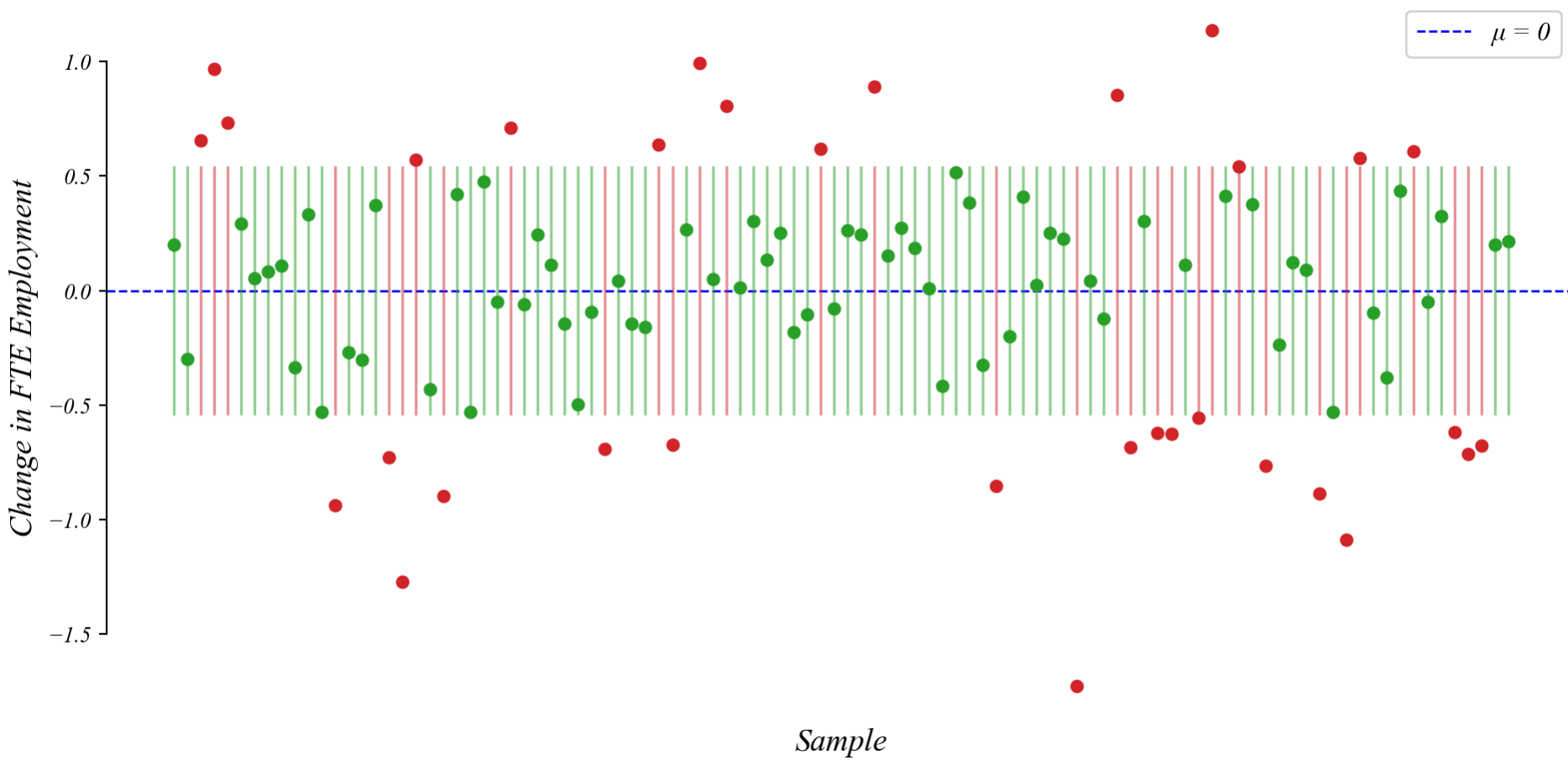
**Question:** what's the probability $\bar{x}$ is closer than $1 \times SE$ to $\mu$?

```
se = sigma / np.sqrt(n)
stats.norm.cdf(mu + 1*se, loc=mu, scale=se) - stats.norm.cdf(mu - 1*se, loc=mu, scale=se)
```

# Confidence Intervals are About Probability

*100 samples, each with ±SE CI centered on $\mu$.*

Each dot is a sample mean $\bar{x}$ from a different sample.




> *the confidence interval around $\mu$ contains most but not all sample means $\bar{x}$*

# Exercise 3.3 | Simulating Confidence Intervals

*Simulate employment changes and calculate the 68% confidence interval.*

## Generate some sample data.

```
sample = np.random.normal(0, 9.8, 331)
```

## Calculate sample statistics.

```
x_bar = np.mean(sample)
mu = 0
se = 9.8 / np.sqrt(331)
mu - se, mu + se
```

```
(-0.5386567157307265, 0.5386567157307265)
```

> *if we took many samples, 68% of the time $\bar{x}$ would land inside this interval*

# The Problem

*We centered the confidence interval on $\mu$.* ***But we don't know*** *$\mu$.*

- *If we knew the true effect of the minimum wage, we wouldn't need the data*
- *So how is a confidence interval centered on $\mu$ useful?*

# The Centerpoint Flip

*The distance between $\bar{x}$ and $\mu$ is the same in both directions.*

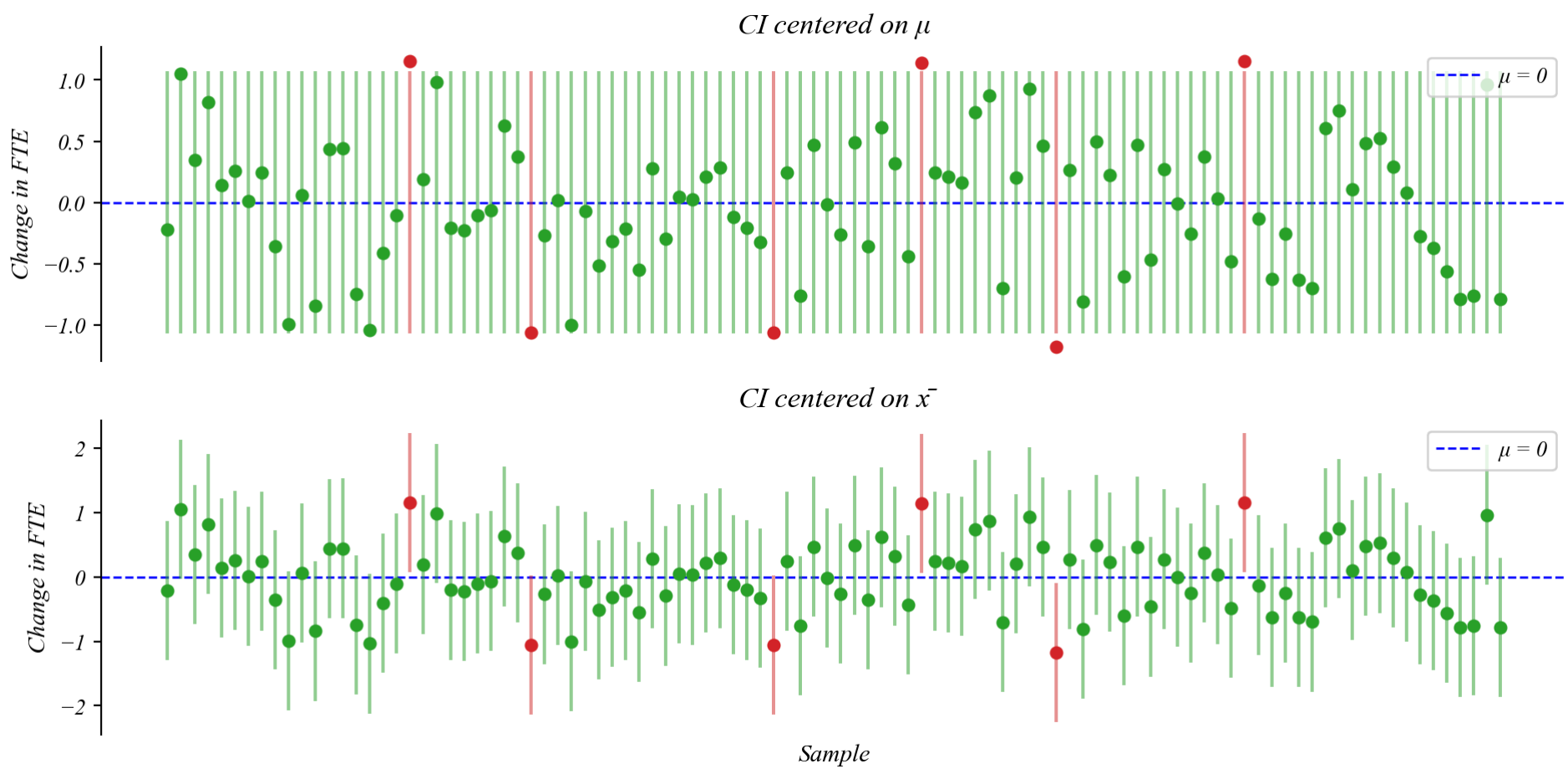
If $\bar{x}$ is within $1 \times SE$ of $\mu$, then $\mu$ is within $1 \times SE$ of $\bar{x}$.

> *instead of asking where $\bar{x}$ lands relative to $\mu$...*

> *we ask where $\mu$ could be relative to our observed $\bar{x}$*

# The Centerpoint Flip

*The distance between $\bar{x}$ and $\mu$ is the same in both directions..*




> *if $\bar{x}$ is in the CI centered on $\mu$, then $\mu$ is in the CI centered on $\bar{x}$*

# New Jersey: 95% Confidence Interval

*Card and Krueger observed $\bar{x} = 0.59$ from one sample of 331 restaurants.*

They drew one sample and centered the CI on $\bar{x}$:

$$\bar{x} \pm 2 \times SE$$

$$= 0.59 \pm 2 \times 0.54$$

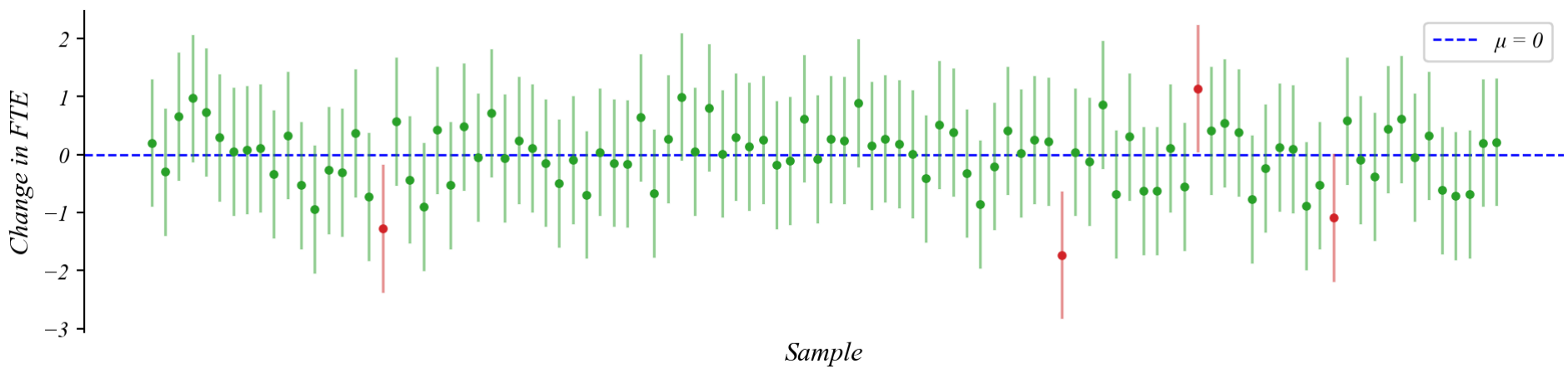
$$= [-0.49,\ 1.67]$$

***Question.*** *Is $\mu$ inside our interval?*

# Confidence Intervals: Interpretation

*Is $\mu$ inside our interval?*

Our study is one of these bars. Which one is it?



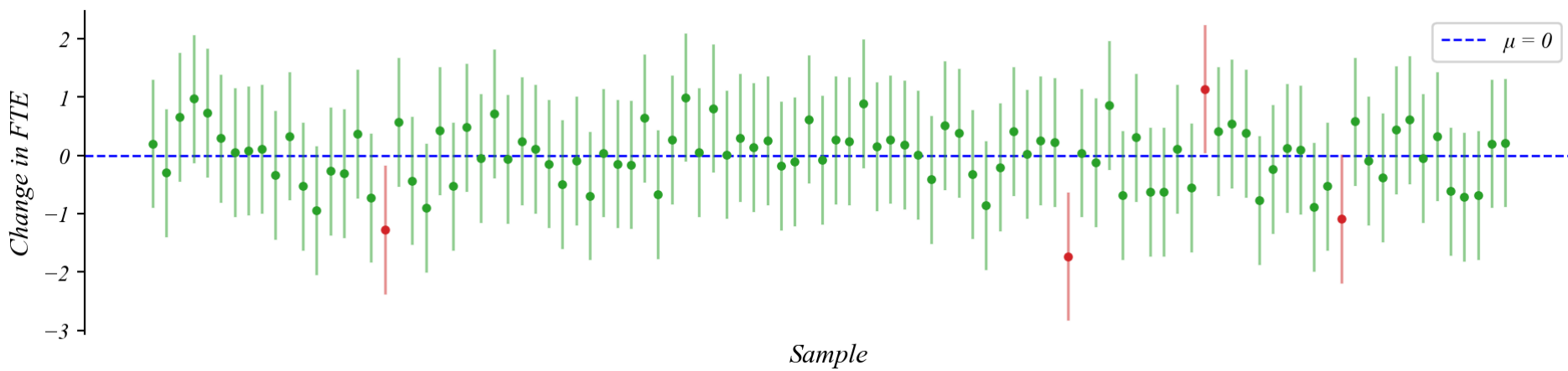

Our interval either contains $\mu$ or it doesn't. We don't know which.

Our "95% confidence" means that **the method works 95% of the time**, not that *this* interval has a 95% chance of containing $\mu$.

# Confidence Intervals: Interpretation

*Is $\mu$ inside our interval?*

Our study is one of these bars. Which one is it?




Our interval either contains $\mu$ or it doesn't. We don't know which.

***Our confidence is in the method, not the specifc interval we estimated!***

# Confidence Intervals: Unknown $\sigma$

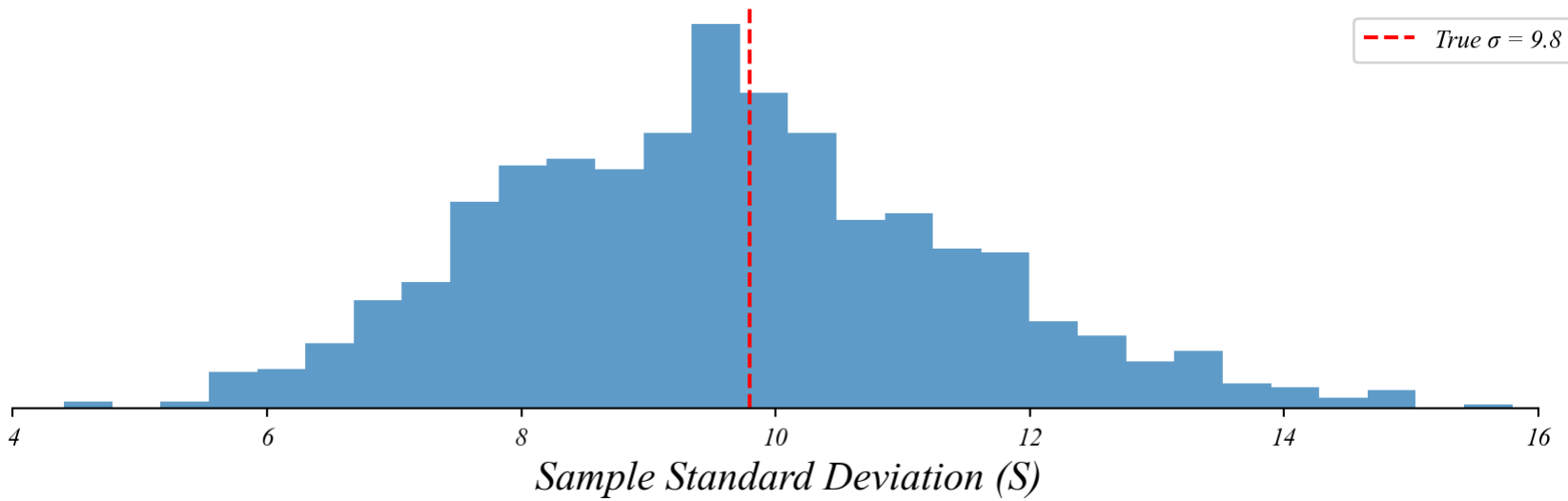
*What if we don't know $\sigma$ either?*

- *We used $\bar{x}$ to estimate $\mu$. Can we use $S$ to estimate $\sigma$?*
- *Yes, but $S$ is itself a random variable, varying from sample to sample.*
- *Replacing $\sigma$ with $S$ introduces a new source of uncertainty.*

# Sampling Variability: $S$

*What if Card and Krueger had only surveyed 15 restaurants?*

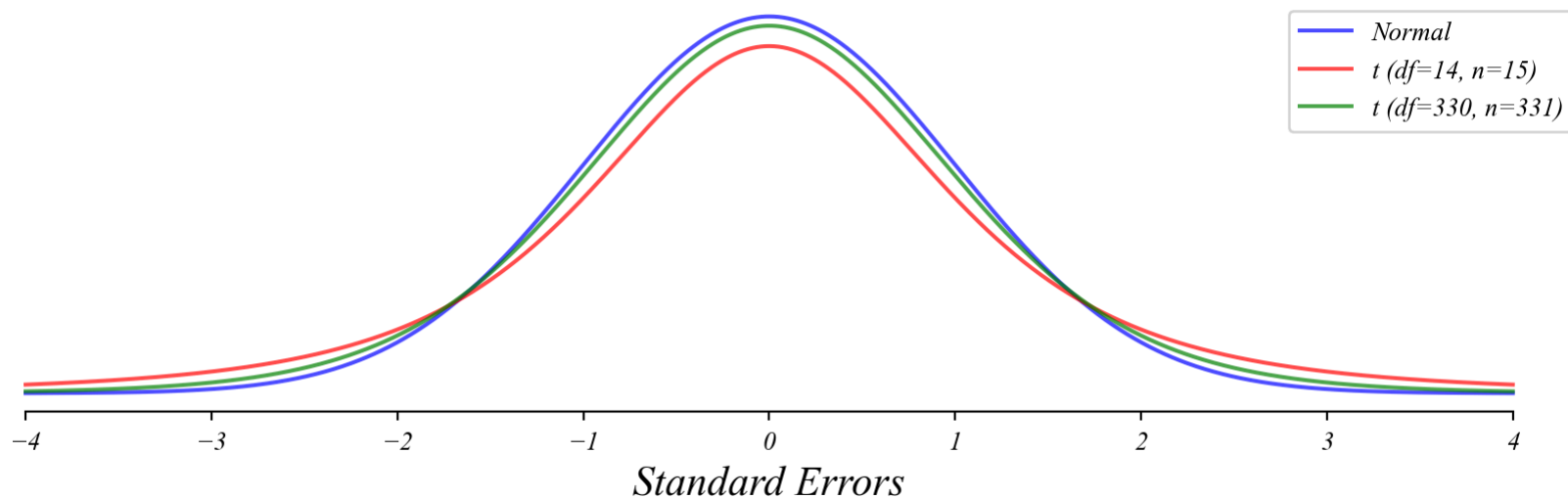
How close would $S$ be to $\sigma$ with only 15 restaurants? Let's draw 1,000 samples.




Like $\bar{x}$ is distributed around $\mu$, $S$ is distributed around $\sigma$.

# The t-Distribution

*...accounts for the extra uncertainty in $S$ around $\sigma$.*




- *The t-Dist accounts for the extra uncertainty when we use $S$ instead of $\sigma$.*
- *When $n$ is large, $S$ is close to $\sigma$ and the t-Dist approaches Normal.*

# Putting It All Together

*Now we can quantify our uncertainty about the true employment effect.*

1. ***Center*** *the sampling distribution on* $\bar{x} = 0.59$.

2. ***Width*** *comes from* $S/\sqrt{n}$ *— the sample standard error.*

3. ***Shape*** *is the t-distribution with* $n - 1$ *degrees of freedom.*

**The Card and Krueger** ($\bar{x} = 0.59$, $S \approx 9.8$, $n = 331$):

$$95\%\ \text{CI}: \quad 0.59 \pm 1.967 \times 0.54 = [-0.47,\ 1.65]$$

# Testing the Theory

*From estimating a range for $\mu$ to testing a specific claim about $\mu$.*

- *Economic theory predicts employment should fall (or at least not change).*
- *Card and Krueger observed $\bar{x} = +0.59$. Is that consistent with no effect?*
- *Instead of asking where $\mu$ might be, we test a specific value: $\mu = 0$.*

# The Null Hypothesis

*We formalize this approach by setting up a "null hypothesis"*

**Null Hypothesis** ($H_0$): *The minimum wage had no effect on employment*

- $H_0 : \mu = 0$

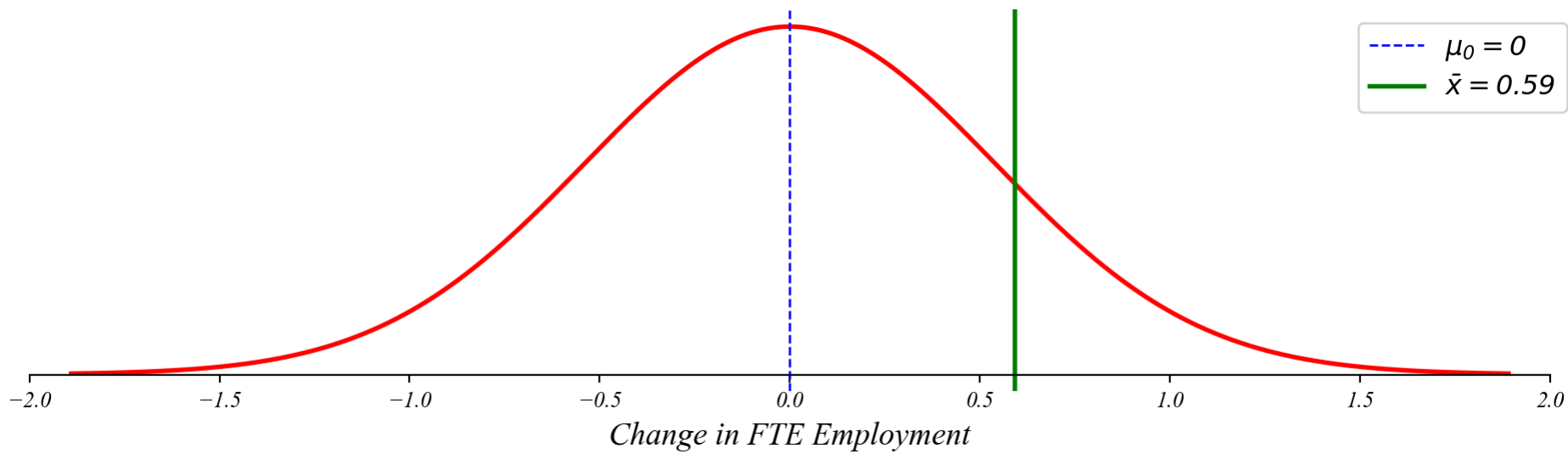
**Testing Question**:

- *How "surprising" is our data if* $H_0$ *were true?*

# NJ Employment Change

*Where does our sample fall on the null distribution?*

If $\mu = 0$, the sampling distribution of $\bar{x}$ is centered on 0.

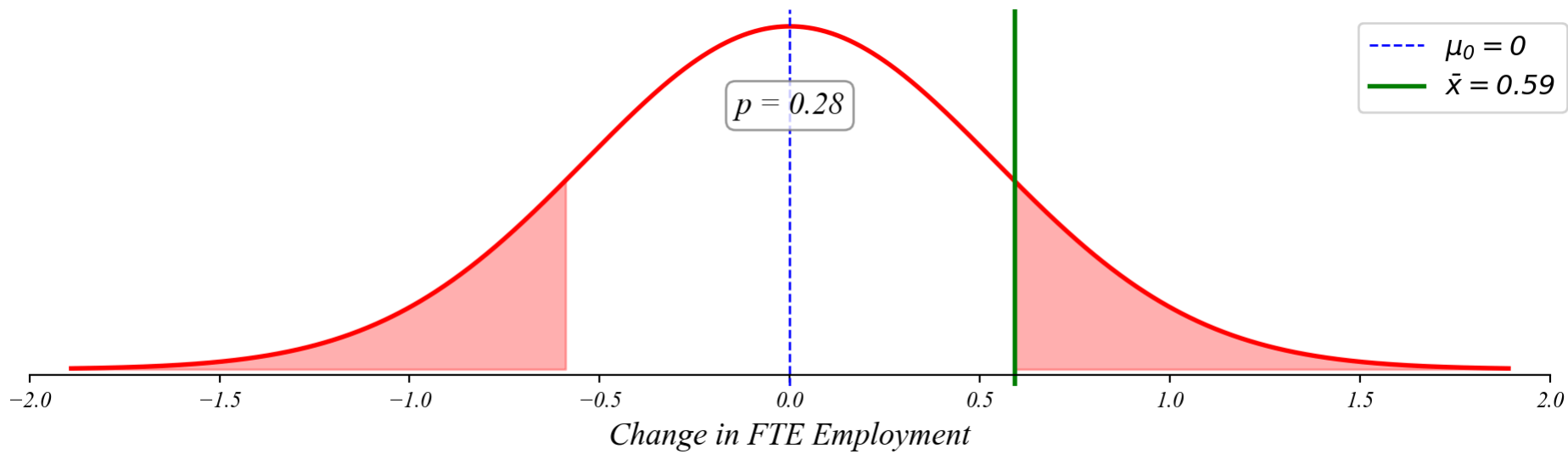



***Questions.*** *Is 0.59 far from 0? How would we know?*

# Quantifying Surprise: The p-value

*What proportion of the null distribution is at least as extreme as our sample?*

If $\mu = 0$, the sampling distribution of $\bar{x}$ is centered on 0.




The shaded area is the probability of getting a result as extreme as ours, if $\mu = 0$.

$p = 0.28$: about 28% of samples would look this extreme or more.

# The t-Statistic

*How many standard errors is our sample mean from the null?*

The **t-statistic** measures how many standard errors our sample mean is from the null value:

$$t = \frac{\bar{x} - \mu_0}{S/\sqrt{n}}$$

$$t = \frac{0.59 - 0}{9.8/\sqrt{331}} = \frac{0.59}{0.54} = 1.09$$

Where:

- $\bar{x}$ *is our sample mean (0.59 FTE)*
- $\mu_0$ *is our null value (0)*
- $S$ *is our sample standard deviation (9.8)*
- $n$ *is our sample size (331)*

# Statistical vs. Practical Significance

*A caution about hypothesis testing*

**Statistical significance:**

- *Formal rejection of the null hypothesis ($p < 0.05$)*
- *Only tells us if the effect is unlikely to be exactly zero*

**Practical significance:**

- *Whether the effect size matters in the real world*
- *Card and Krueger: the direction contradicted theory, even though NJ alone wasn't significant*

> *always consider the magnitude of the effect, not just the p-value*

# Looking Ahead

*The hypothesis testing framework is powerful and general.*

- *Part 3.4 |* ***The Simplest Linear Model***
- *Part 4 |* ***Bivariate Models***

*> the tools we built today apply to every model we'll see in Parts 4 and 5*